THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 02

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Which pair of quantities is represented by the symbols m and v in science? 1

(a) mass and velocity **(b)** metre and volume
(c) moles and voltage **(d)** mirror and vision

2. The compass on the page numbers of the textbook stands for: 1

(a) direction — choosing models and the right questions
(b) memorising laws
(c) careful observation only
(d) taking readings

3. Particles in a true solution do NOT scatter light because: 1

(a) they are charged **(b)** they are too small
(c) they are too large **(d)** they are coloured

4. Which method separates two miscible liquids with boiling points differing by more than about 25 °C? 1

(a) sublimation **(b)** separating funnel
(c) simple distillation **(d)** filtration

5. Which postulate of Dalton's atomic theory is correct? 1

(a) Atoms can be divided in a chemical reaction
(b) Atoms of different elements are identical
(c) Atoms cannot be created or destroyed in a chemical reaction
(d) Atoms combine in any ratio

6. Which of the following is a typical property of ionic compounds? 1

(a) low melting point **(b)** high melting and boiling points
(c) soft and flammable **(d)** poor solubility in water always

7. Rutherford's model could NOT explain: 1

(a) scattering of α-particles **(b)** the stability of the atom
(c) the existence of a nucleus **(d)** the neutrality of atoms

8. A body thrown vertically upward and falling back to the thrower's hand has displacement: 1

(a) the maximum height **(b)** equal to the distance
(c) zero **(d)** twice the height

9. A body in uniform circular motion of radius r completes one revolution in time T. Its average speed is: 1

(a) πr/T **(b)** r/T **(c)** 2πr/T **(d)** zero

10. A body of mass 2 kg is acted on by two forces, 8 N east and 3 N west. Its acceleration is: 1

(a) 1.5 m s^{-2} east **(b)** 2.5 m s^{-2} west **(c)** 5.5 m s^{-2} east **(d)** 2.5 m s^{-2} east

11. A body of mass 5 kg initially at rest is pushed by a net force of 10 N for 3 s. Its final velocity is: 1

(a) 6 m s^{-1} **(b)** 30 m s^{-1} **(c)** 3 m s^{-1} **(d)** 12 m s^{-1}

12. The kinetic energy of a body of mass 4 kg moving at 5 m s^{-1} is: 1

(a) 100 J **(b)** 25 J **(c)** 50 J **(d)** 20 J

13. A source of frequency 2000 Hz produces waves of wavelength 0.17 m in a medium. The speed of the wave is: 1

(a) 11764.71 m s^{-1} **(b)** 340 m s^{-1} **(c)** 2000.17 m s^{-1} **(d)** 680 m s^{-1}

14. Protein synthesis takes place on: 1

(a) lysosomes **(b)** vacuoles **(c)** Golgi bodies **(d)** ribosomes

15. The basic structural and functional unit of life is the: 1

(a) tissue **(b)** cell **(c)** organ **(d)** nucleus

16. The tissue that provides cushioning at joints and in the nose and ear is: 1

(a) adipose **(b)** bone **(c)** cartilage **(d)** blood

17. The basic unit of classification is the: 1

(a) species **(b)** family **(c)** order **(d)** genus

18. Internal fertilisation takes place in: 1

(a) frogs only **(b)** fishes only
(c) sponges **(d)** reptiles, birds and mammals

19. Fungi are the source of the antibiotic: 1

(a) penicillin **(b)** paracetamol **(c)** insulin **(d)** vitamin C

20. Chemicals that caused the ozone hole are: 1

(a) CFCs **(b)** nitrogen oxides only
(c) methane **(d)** CO_2

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. What should scientists do when a prediction does not match the observation? 2

22. How is a covalent bond formed? Give one example. 2

23. What is the direction of velocity in uniform circular motion? 2

24. Calculate the weight of a 30 kg body (g = 10 m s^{-2}). 2

25. Why is the nucleus called the control centre of the cell? 2

26. Why is sclerenchyma strong? 2

27. State two functions of the placenta. 2

28. Why does wind blow? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. What are the differences between evaporation and crystallization? State which gives pure crystals. 3

30. Why is the weighted average atomic mass of chlorine 35.5 u even though no atom has this mass? 3

31. On a seesaw a 15 kg child sits 2 m from the fulcrum. Where must a 30 kg child sit to balance it? 3

32. Describe how bats navigate in the dark. 3

33. What are the components of blood? State its function. 3

34. Why is the Western Ghats region protected as a biodiversity hotspot and how do sacred groves help? 3

35. Write the names of: 3

(a) CS_2
(b) N_2O_4
(c) PCl_3
(d) H_2S
(e) NH_3 (common name). State the rule for prefixes.

OR

Explain the formation of sodium chloride from its atoms. 3

36. What is Mission LiFE? 3

OR

Explain how deforestation affects the water cycle. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Write the chemical formulae of calcium chloride, aluminium oxide, sodium sulfate, magnesium oxide and ammonia. 5
(ii) Name the rule used.
(iii) Name CO, CO_2, CS_2, N_2O_4, PCl_3.

OR

(i) What is a molecule? 5
(ii) Distinguish between a molecule of an element and of a compound with two examples each.
(iii) Why does helium exist as single atoms?

38. (i) Differentiate mass and weight. 5
(ii) What is the weight of a 60 kg person on the Earth (g = 10)?
(iii) On the Moon where g = 1.6?

OR

Case study: A weightlifter holds a barbell of total mass 30 kg (g = 9.8 m s^{-2}). 5

(i) What forces act on it?

(ii) Find the weight.

(iii) Find the force she applies.

(iv) What would be the net force?

39. (i) How does the structure of a red blood cell suit its function? 5

(ii) Give another example of a specialised cell.

(iii) What is the length of an RBC lifespan?

OR

Case study: A student wants to estimate the size of cells under the microscope: a field of 4 mm shows 20 cells in a row; eyepiece 10X and objective 10X. 5

(i) Find the size of one cell.

(ii) What is the total magnification?

(iii) How big does one cell appear?

(iv) Which part of the microscope gives clarity?

40. (i) Describe the human male and female reproductive systems. 5

(ii) State the functions of the testes and ovaries.

OR

Case study: Ravi is growing taller, his voice is changing and his shoulders broaden. 5

(i) Which stage is this?

(ii) What brings about these changes?

(iii) Mention two changes in girls.

(iv) Why is a balanced diet important?

END OF QUESTION PAPER **Total: 80 Marks**

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